

Gaussian NLL derivation

In the Bernoulli case, we derived the log-likelihood by the logistic $\sigma(a)$ then applied the chain rule w.r.t. the logit $a = wX + b$.

Here we want:

$$\leadsto a_{\mu} = \underline{w_{\mu}}X + \underline{b_{\mu}} \quad \leadsto a_{\sigma^2} = \text{softplus}(\underline{w_{\sigma^2}}X + \underline{b_{\sigma^2}})$$

$$p(y|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\left(-\frac{(y-\mu)^2}{2\pi\sigma^2}\right)}$$

$$= -\log((2\pi\sigma^2)^{1/2}) - \frac{(y-\mu)^2}{2\sigma^2} = \underbrace{L}_{\text{L}} = + \frac{\log(2\pi\sigma^2)}{2} + \frac{(y-\mu)^2}{2\sigma^2}$$

We first derive by the predicted $\hat{\mu} = w_{\mu}X + b_{\mu}$, $\hat{\sigma}^2 = \text{softplus}(w_{\sigma^2}X + b_{\sigma^2})$ then apply the chain rule w.r.t. the logit.

$$\leadsto \frac{\partial L}{\partial \mu} = \frac{(y-\mu)}{\sigma^2}$$

$$\leadsto \frac{\partial L}{\partial \sigma^2} = \frac{\log(2\pi)}{2} + \log(\sigma) + \frac{(y-\mu)^2}{2\sigma^2} = \frac{1}{\hat{\sigma}} - (y-\hat{\mu})^2 \cdot (\hat{\sigma}^{-3})$$

chain rule:

$$\leadsto \frac{\partial L}{\partial \mu} = \frac{\partial L}{\partial \mu} \cdot \frac{\partial \mu}{\partial a_{\mu}} = \left(\frac{y-\mu}{\sigma^2} \right)$$

$$\leadsto \frac{\partial L}{\partial \sigma^2} = \frac{\partial L}{\partial \sigma^2} \cdot \frac{\partial \sigma^2}{\partial a_{\sigma^2}} = \left(\frac{1}{\hat{\sigma}} - (y-\hat{\mu})^2 \cdot (\hat{\sigma}^{-3}) \right) \cdot \frac{\partial \text{softplus}(a_{\sigma^2})}{\partial a_{\sigma^2}}$$

$$\frac{\partial \text{softplus}(a_{\sigma^2})}{\partial a_{\sigma^2}} = \partial \left\{ \frac{e^{a_{\sigma^2}}}{1 + e^{a_{\sigma^2}}} \right\} / da_{\sigma^2} = \sigma(a) \leadsto \text{sigmoid}$$

$$\leadsto \frac{\partial L}{\partial \mu} = \left(\frac{y-\mu}{\sigma^2} \right)$$

$$\leadsto \frac{\partial L}{\partial \sigma^2} = \left(\frac{1}{\hat{\sigma}} - (y-\hat{\mu})^2 \cdot (\hat{\sigma}^{-3}) \right) \cdot \sigma(a_{\sigma^2})$$

$$\left\{ \begin{array}{l} \frac{\partial L}{\partial w_{\mu}} = \frac{\partial L}{\partial a_{\mu}} \cdot \frac{\partial a_{\mu}}{\partial w_{\mu}} = \left(\frac{y - \mu}{\sigma^2} \right) \cdot X \\ \frac{\partial L}{\partial b_{\mu}} = \frac{\partial L}{\partial a_{\mu}} \cdot \frac{\partial a_{\mu}}{\partial b_{\mu}} = \left(\frac{y - \mu}{\sigma^2} \right) \end{array} \right.$$

$$\left\{ \begin{array}{l} \frac{\partial L}{\partial w_{\sigma^2}} = \frac{\partial L}{\partial a_{\sigma^2}} \cdot \frac{\partial a_{\sigma^2}}{\partial w_{\sigma^2}} = \left(\frac{1}{\hat{\sigma}} - (y - \hat{\mu})^2 \cdot (\hat{\sigma}^{-3}) \right) \cdot \sigma(a_{\sigma^2}) \cdot X \\ \frac{\partial L}{\partial b_{\mu}} = \frac{\partial L}{\partial a_{\mu}} \cdot \frac{\partial a_{\mu}}{\partial b_{\mu}} = \left(\frac{1}{\hat{\sigma}} - (y - \hat{\mu})^2 \cdot (\hat{\sigma}^{-3}) \right) \cdot \sigma(a_{\sigma^2}) \end{array} \right.$$